Lecture 25

Section 5.4

Eigenvectors and Linear Transformations

Idea: We have discussed the notion of eigenvalues and eigenvectors of a matrix. We would like to generalize this idea to linear transformations.

Definition: Suppose $T : V \to V$ is a linear transformation. An eigenvector of T is a nonzero vector $\mathbf{v}$ where $T(\mathbf{v}) = \lambda\mathbf{v}$ for some scalar λ. An eigenvalue of T is a scalar λ such that there exists a nonzero vector $\mathbf{v}$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$. We say "$\mathbf{v}$ corresponds to the eigenvalue λ."

Let V be an n-dimensional vector space. Let $T : V \to V$ be a linear transformation.

Goal: Associate a matrix to T.

Choose any basis $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ for V. Suppose $\mathbf{v} \in V$. Then,

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n.$$

Note that this expression is unique, since $\mathcal{B}$ is a basis. The coordinate vector of $\mathbf{v}$ with respect to $\mathcal{B}$, denoted by $[\mathbf{v}]_{\mathcal{B}}$, is

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n.$$

Notice that

$$\begin{aligned} T(\mathbf{v}) &= T(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n) \\ &= a_1T(\mathbf{v}_1) + a_2T(\mathbf{v}_2) + \cdots + a_nT(\mathbf{v}_n). \end{aligned}$$

Because the coordinate mapping from V to $\mathbb{R}^n$ is linear, this implies that

$$[T(\mathbf{v})]_{\mathcal{B}} = a_1[T(\mathbf{v}_1)]_{\mathcal{B}} + a_2[T(\mathbf{v}_2)]_{\mathcal{B}} + \cdots + a_n[T(\mathbf{v}_n)]_{\mathcal{B}}.$$

Rewriting the right side of this equation, it follows that

$$[T(\mathbf{v})]_{\mathcal{B}} = M[\mathbf{v}]_{\mathcal{B}}$$

where

$$M = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ [T(\mathbf{v}_1)]_{\mathcal{B}} & [T(\mathbf{v}_2)]_{\mathcal{B}} & \cdots & [T(\mathbf{v}_n)]_{\mathcal{B}} \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}.$$

We say that M is the matrix for T relative to the basis $\mathcal{B}$ (or the $\mathcal{B}$-matrix for T), denoted by $[T]_{\mathcal{B}}$.

Takeaway: The action of T on $\mathbf{v}$ can be viewed as left-multiplication by $[T]_{\mathcal{B}}$ on $[\mathbf{v}]_{\mathcal{B}}$.

$$\begin{tikzcd}
\mathbf{v} \arrow[r, "T"] \arrow[d, dashed] & T(\mathbf{v}) \arrow[d, dashed] \\
{[\mathbf{v}]_{\mathcal{B}}} \arrow[r, "{[T]_{\mathcal{B}}}"] & {[T(\mathbf{v})]_{\mathcal{B}}}
\end{tikzcd}$$

Example: Consider $T : \mathbb{P}_2 \to \mathbb{P}_2$ where $T(p(t)) = p'(t)$. Suppose $\mathcal{B} = \{1, t, t^2\}$ is the basis for $\mathbb{P}_2$. Find the $\mathcal{B}$-matrix for T.

Solution: Notice that

$$T(1) = 0, \qquad T(t) = 1, \qquad T(t^2) = 2t.$$

Then,

$$[T(1)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad [T(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad [T(t^2)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

So,

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

We see that

$$\begin{tikzcd}
a + bt + ct^2 \arrow[r, "\text{act by } T"] \arrow[d, dashed] & b + 2ct \arrow[d, dashed] \\
\begin{bmatrix} a \\ b \\ c \end{bmatrix} \arrow[r, "\text{multiplication by } [T]_{\mathcal{B}}"] & \begin{bmatrix} b \\ 2c \\ 0 \end{bmatrix}
\end{tikzcd}.$$

Theorem (Diagonal Matrix Representation): Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed from the columns of P (i.e. the eigenvectors of A), then $D = [T]_{\mathcal{B}}$, where $T(\mathbf{v}) = A\mathbf{v}$.

$$\begin{tikzcd}
\mathbf{v} \arrow[r, "\text{multiplication by } A"] \arrow[d, dashed, "\text{multiplication by } P^{-1}"'] & A\mathbf{v} = PDP^{-1}\mathbf{v} \\
{[\mathbf{v}]_{\mathcal{B}} = P^{-1}\mathbf{v}} \arrow[r, "\text{multiplication by } D"] & {[A\mathbf{v}]_{\mathcal{B}} = DP^{-1}\mathbf{v}} \arrow[u, dashed, "\text{multiplication by } P"']
\end{tikzcd}$$

Note: Both A and D describe the same linear transformation, relative to different bases.

Theorem: The eigenvalues of $T : V \to V$ are exactly the eigenvalues of $[T]_{\mathcal{B}}$, where $\mathcal{B}$ is any basis of V. Moreover, the eigenvectors of $[T]_{\mathcal{B}}$ are the coordinate vectors of the eigenvectors of T.

Example: Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{v}) = A\mathbf{v}$, where

$$A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}.$$

Find a basis $\mathcal{B}$ for $\mathbb{R}^2$ with the property that $[T]_{\mathcal{B}}$ is diagonal.

Solution: In lecture 24, we found a diagonal matrix D and invertible matrix P such that $A = PDP^{-1}$, where

$$D = \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \qquad \text{and} \qquad P = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}.$$

We did this by finding the eigenvalues and eigenvectors of A. So, we can let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$.

Remark: The proof of the diagonal matrix theorem does not rely on D being diagonal. So, if A is similar to a matrix C, where $A = PCP^{-1}$, then $C = [T]_{\mathcal{B}}$ is the $\mathcal{B}$-matrix for the transformation $T(\mathbf{v}) = A\mathbf{v}$, where the basis $\mathcal{B}$ is formed from the columns of P.

Example: Find the $\mathcal{B}$-matrix for the transformation $T(\mathbf{v}) = A\mathbf{v}$, where $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and

$$A = \begin{bmatrix} -5 & -1 \\ 6 & -1 \end{bmatrix}, \qquad \mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{b}_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

Solution: Using the above remark, we want to find $C = P^{-1}AP$. We let $P = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix}$. So,

$$P = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}, \qquad P^{-1} = \frac{1}{-1-(-2)} \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}.$$

We compute

$$\begin{aligned} C &= \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -5 & -1 \\ 6 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -7 & 6 \\ 4 & -5 \end{bmatrix} \\ &= \begin{bmatrix} 11 & -11 \\ 18 & -17 \end{bmatrix}, \end{aligned}$$

and so $C = [T]_{\mathcal{B}}$.